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A Supplementary Results

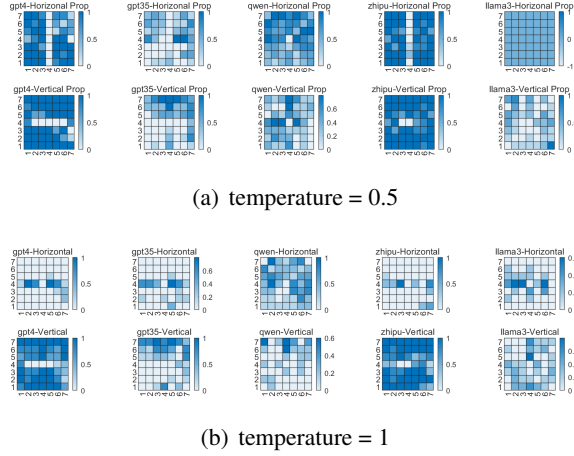


Figure S1: Performance of five LLMs (i.e. GPT-4, GPT-3.5-Turbo, Qwen-Turbo, ZhipuAI, and Llama3-8B) on spatial preposition preference at all cells except the center ([4,4]).

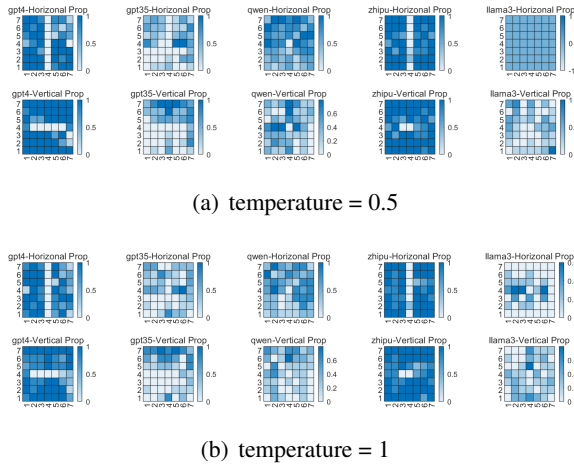


Figure S2: Distribution of horizontal and vertical spatial prepositions appeared in LLMs' responses in the spatial generation task.

B Prompts for Spatial Representation Tasks

Prompt templates for the spatial generation task and the spatial rating task are provided below.

1) **Spatial Generation Task:** "On a 7*7 grid, the bottom left corner is [1,1], while the top right corner is [7,7]. The [figure] is at [x1, y1], while the [reference] is at [4,4]. So, the [figure] is [relation] the [reference]. Please give appropriate spatial prepositions to replace the [relation]. Avoid using compass directions, a clock face, or the degree of angle."

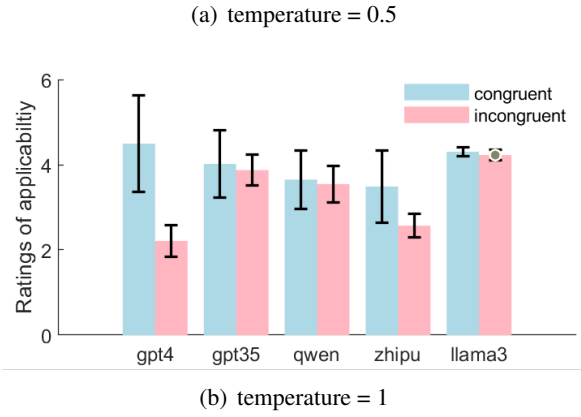
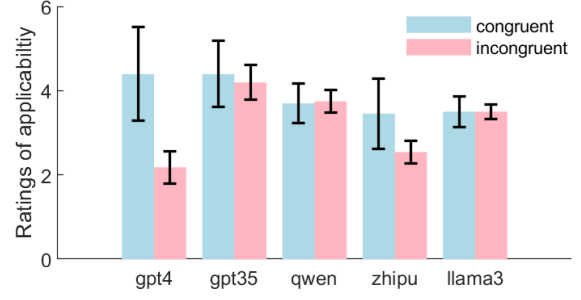
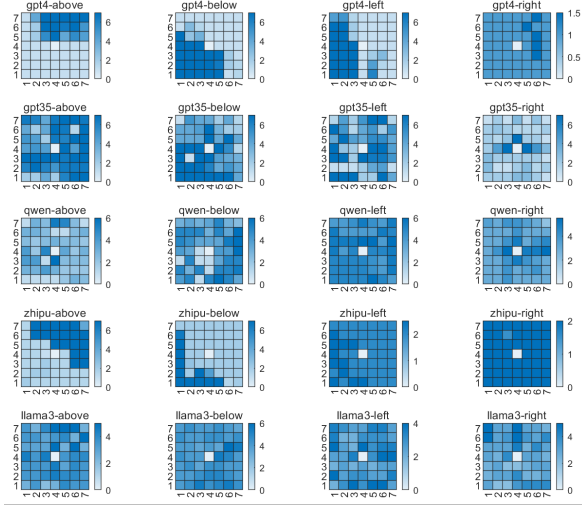


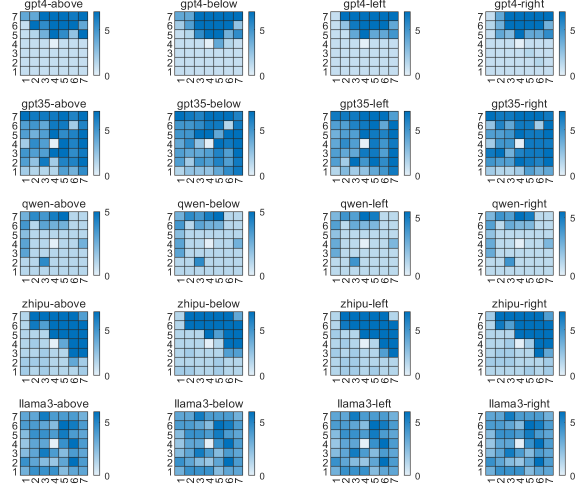
Figure S3: Performance of five LLMs on the spatial rating task. LLMs' ratings are compared between the congruent and incongruent conditions where the descriptions of spatial relations between the figure object and the reference object either correspond to the truth or not.

2) **Spatial Rating Task:** "On a 7*7 grid, the bottom left corner is [1,1], while the top right corner is [7,7]. The [figure] is at [x1, y1], while the [reference] is at [4,4]. Please rate the appropriateness of the following statement on a scale of 1 to 7, where 1 is the least appropriate and 7 is the most appropriate. The Statement is: The [figure] is [relation] the [reference]."

The specific prompt with placeholders replaced by actual items is available on this anonymous website [Spatial Representations of LLMs](#).

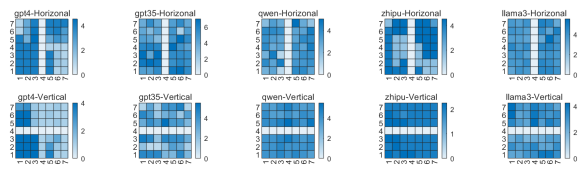


(a) temperature = 0.5



(b) temperature = 1

Figure S4: Performance of five LLMs on the spatial rating task.



(a) temperature = 0.5



(b) temperature = 1

Figure S5: LLMs' ratings on each location where the figure object is situated at around the reference object.